

Manipulating Data

1. Create EmpTest from Emp table by copying structure and data.

```
SQL> CREATE TABLE EmpTest11 AS SELECT * FROM Emp;
```

```
SQL> CREATE TABLE EmpTest11 AS SELECT * FROM Emp;
Table created.
```

1. Add a new row into the EmpTest table for Empno, Ename, Sal columns. In Ename should have the Current User.

```
SQL> INSERT INTO EmpTest 11(Empno, Ename, Sal) VALUES (1234, 'Sowmya', 5000);
```

```
SQL> INSERT INTO EmpTest11 (Empno, Ename, Sal)
  2  VALUES (1234, 'Sowmya', 5000);
1 row created.
```

2. Update EmpTest by increasing the Salary of TURNER by 15%. Confirm your changes.

```
SQL> UPDATE EmpTest11
```

```
SET Sal = Sal* 1.15
```

```
WHERE Ename = 'TURNER';
```

```
SQL> UPDATE EmpTest11 SET Sal=Sal*1.5 where Ename='turner';
0 rows updated.
```

```
SQL> SELECT * FROM EmpTest11
```

```
WHERE Ename='TURNER';
```

```
SQL> SELECT * FROM EmpTest11 WHERE Ename='TURNER';
```

EMPNO	ENAME	JOB	MGR	HIREDATE
103	TURNER	SALESMAN	105	08-SEP-81
1500	0	30		

3. Update the salary of Smith with salary of Scott using EmpTest table.

```
SQL> UPDATE EmpTest SET Salary = (SELECT Salary FROM EmpTest
WHERE Ename = 'SCOTT') WHERE Ename = 'SMITH';
```

```
SQL> UPDATE EmpTest SET Sal=(SELECT Sal FROM EmpTest WHERE Ename='SCOTT') WH
ERE Ename='SMITH';

9 rows updated.
```

4. Increase all the employees salary by 10% in EmpTest table who are working in NEW YORK.

```
SQL> UPDATE EmpTest11 SET Sal= Sal* 1.10 WHERE Deptno IN
( SELECT Deptno FROM Dept WHERE Loc='NEW YORK' );
```

```
SQL> UPDATE EmpTest11
2 SET Sal = Sal * 1.10
3 WHERE Deptno IN (
4 SELECT Deptno
5 FROM Dept
6 WHERE Loc = 'NEW YORK'
7 );

4 rows updated.
```

5. Delete all the Comm data from EmpTest Table;

```
SQL>UPDATE EmpTest SET Comm = NULL;
```

```
SQL> UPDATE EmpTest11 SET Comm = NULL;

16 rows updated.
```

6. Delete all the employees from EmpTest table who are working in SALES dept.

```
SQL> DELETE FROM EmpTest WHERE Deptno =
( SELECT Deptno
FROM Dept
WHERE Dname='SALES'
);
```

```
SQL> DELETE FROM EmpTest11 WHERE Deptno=(SELECT Deptno FROM Dept WHERE Dname='SALES');

6 rows deleted.
```

7. Delete all the who are working with that employee, except that Employee.(Prompt for the ENAME).

```
SQL> DELETE FROM EmpTest
WHERE Deptno =
(SELECT Deptno
FROM EmpTest
```

WHERE Ename='&ENAME') AND Ename<>'&ENAME';

```
SQL> DELETE FROM EmpTest11 WHERE Deptno=(SELECT Deptno FROM EmpTest11 WHERE Ename='&ENAME') AND Ename<>'&ENAME';
Enter value for ename: SMITH
Enter value for ename: SMITH
old 1: DELETE FROM EmpTest11 WHERE Deptno=(SELECT Deptno FROM EmpTest11 WHERE Ename='&ENAME') AND Ename<>'&ENAME'
new 1: DELETE FROM EmpTest11 WHERE Deptno=(SELECT Deptno FROM EmpTest11 WHERE Ename='SMITH') AND Ename<>'SMITH'
0 rows deleted.
```

8. Create Emp2 from Emp by only copying Empno,Ename,sal without copying data.

```
SQL> CREATE TABLE Emp2 AS
      SELECT Empno,Ename,Sal
      FROM Emp
      WHERE 1=2;
```

```
SQL>  CREATE TABLE Emp2 AS
      2      SELECT Empno,Ename,Sal
      3      FROM Emp
      4      WHERE 1=2;
Table created.
```

9. Create Emp3 from Emp by only copying Empno,Job without copying data.

```
SQL> CREATE TABLE Emp3 AS
      SELECT Empno,Job
      FROM Emp
      WHERE 1=2;
```

```
SQL> CREATE TABLE Emp3 AS
      2      SELECT Empno,Job
      3      FROM Emp
      4      WHERE 1=2;
Table created.
```

10. Using multitable insert Emp data into Emp2 and Emp3 Tables.

```
SQL> INSERT ALL
      INTO Emp2(Empno,Ename,Sal)
      VALUES(Empno,Ename,Sal)
      INTO Emp3(Empno,Job)
      VALUES(Empno,Job)
      SELECT Empno,Ename,Sal,Job
      FROM Emp;
```

```
SQL> INSERT ALL
  2          INTO Emp2(Empno,Ename,Sal)
  3          VALUES(Empno,Ename,Sal)
  4          INTO Emp3(Empno,Job)
  5          VALUES(Empno,Job)
  6          SELECT Empno,Ename,Sal,Job
  7          FROM Emp;

30 rows created.
```

11. Truncate Emp2 Table and insert following two rows as follows: 7788, SMITH, 4500 /7654, JACK, 3500.

SQL> TRUNCATE TABLE Emp2;

```
SQL> TRUNCATE TABLE Emp2;

Table truncated.
```

SQL>INSERT INTO Emp2 VALUES(7788,'SMITH',4500);

```
SQL> INSERT INTO Emp2 VALUES(7788,'SMITH',4500);

1 row created.
```

SQL>INSERT INTO Emp2 VALUES(7654,'JACK',3500);

```
SQL> INSERT INTO Emp2 VALUES(7654,'JACK',3500);

1 row created.
```

12. Commit the Data.

SQL>commit;

```
SQL> commit;

Commit complete.
```

13. Using Merge statement insert and update Emp2 using Emp.

SQL> MERGE INTO Emp2 E2

USING Emp E

ON (E2.Empno=E.Empno)

WHEN MATCHED THEN

UPDATE SET

E2.Ename=E.Ename,

E2.Sal=E.Sal

WHEN NOT MATCHED THEN

```
INSERT(Empno,ENAME,SAL)
VALUES(E.Empno,E.ENAME,E.SAL);
```

```
SQL> MERGE INTO Emp2 E2
  2 USING Emp E
  3 ON (E2.Empno=E.Empno)
  4 WHEN MATCHED THEN
  5 UPDATE SET
  6 E2.ENAME=E.ENAME,
  7 E2.SAL=E.SAL
  8 WHEN NOT MATCHED THEN
  9 INSERT(Empno,ENAME,SAL)
 10 VALUES(E.Empno,E.ENAME,E.SAL);

15 rows merged.
```

14. Verify your changes.

```
SQL>select * from emp2;
```

```
SQL> select * from emp2;

  EMPNO  ENAME          SAL
-----
  7788 SMITH             4500
  7654 JACK             3500
   113 ADAMSSMITH       1100
   107 ALLEN            1600
   108 WARDEN           1250
   109 MARTINKING       1250
   105 BLAKE            2850
   110 JAMES             950
   104 SMITHKO           800
   115 RAMESH           2600
   101 KING             5000

  EMPNO  ENAME          SAL
-----
   103 TURNER           1500
   112 CLARKEN          2450
   111 MILLERROBIN      1300
   114 SARALA           2100
   102 SCOTLER          3000
   106 JONES            2975

17 rows selected.
```

15. Rollback the Data.

```
SQL>rollback;
```

```
SQL> rollback;

Rollback complete.
```

17.Using Merge statements update Emp2 table for only Empno=7788 and Insert only those employees whose salary is more than 3000.

```
SQL> MERGE INTO Emp2 E2
USING
(
SELECT *
FROM Emp
WHERE Sal>3000
) E
ON(E2.Empno=E.Empno)
WHEN MATCHED THEN
UPDATE SET
E2.Ename=E.Ename,
E2.Sal=E.Sal
WHERE E.Empno=7788
WHEN NOT MATCHED THEN
INSERT(Empno,Ename,Sal)
VALUES(E.Empno,E.Ename,E.Sal);
```

```
SQL> MERGE INTO Emp2 E2
2 USING
3 (
4 SELECT *
5 FROM Emp
6 WHERE Sal>3000
7 ) E
8 ON(E2.Empno=E.Empno)
9 WHEN MATCHED THEN
10 UPDATE SET
11 E2.Ename=E.Ename,
12 E2.Sal=E.Sal
13 WHERE E.Empno=7788
14 WHEN NOT MATCHED THEN
15 INSERT(Empno,Ename,Sal)
16 VALUES(E.Empno,E.Ename,E.Sal);
1 row merged.
```

18.Verify your changes.

```
SQL>select * from emp2;
```

```
SQL> select * from emp2;
```

EMPNO	ENAME	SAL
7788	SMITH	4500
7654	JACK	3500
101	KING	5000

19.Create a User WIPRO.

SQL> CREATE USER WIPRO IDENTIFIED BY wipro;

```
SQL> CREATE USER WIPRO IDENTIFIED BY wipro;
User created.
```

SQL> GRANT CONNECT, RESOURCE TO WIPRO;

```
SQL> GRANT CONNECT, RESOURCE TO WIPRO;
Grant succeeded.
```

20. Grant ALL permission on EMP table from SCOTT to WIPRO user.

SQL> GRANT ALL ON Emp TO WIPRO;

```
SQL> GRANT ALL ON EMP TO WIPRO;
Grant succeeded.
```

21. Delete all the employees in deptno 10 and do not issue a Commit.

Delete all the employees in deptno 10 and do not issue a Commit.

SQL> DELETE FROM Emp WHERE Deptno=10;

```
SQL> DELETE FROM Emp WHERE Deptno=10;
4 rows deleted.
```

22. From WIPRO user delete all the employees from SCOTT.EMP table in DEPTNO=10. What happens and Why?

SQL> DELETE FROM Scott.Emp WHERE Deptno=10;

```
SQL> DELETE FROM SCOTT.EMP WHERE DEPTNO = 10;
4 rows deleted.
```

23. Issue a Rollback in SCOTT user and Check the WIPRO user.

SQL> ROLLBACK;

```
SQL> ROLLBACK;
Rollback complete.
```

SQL> SELECT * FROM Scott.Emp WHERE Deptno=10;

```
SQL> SELECT * FROM Scott.Emp WHERE Deptno=10;

  EMPNO  ENAME          JOB               MGR HIREDATE
-----
    SAL      COMM      DEPTNO
-----
    101 KING            PRESIDENT          17-NOV-81
    5000              10
    111 MILLERROBIN      CLERK              101 23-JAN-82
    1300              10
    112 CLARKEN          MANAGER            101 09-JUN-81
    2450              10

  EMPNO  ENAME          JOB               MGR HIREDATE
-----
    SAL      COMM      DEPTNO
-----
    114 SARALA          CLERK              112 15-AUG-81
    2100              10
```

24. In SCOTT user give a query on EMP using FOR UPDATE clause with WAIT 20 seconds? What happens?

SQL> SELECT * FROM Emp FOR UPDATE WAIT 20;

```
SQL> SELECT * FROM Emp FOR UPDATE WAIT 20;

  EMPNO  ENAME          JOB               MGR HIREDATE
-----
    SAL      COMM      DEPTNO
-----
    113 ADAMSSMITH          CLERK              102 23-MAY-87
    1100              20
    101 KING            PRESIDENT          17-NOV-81
    5000              10
    102 SCOTLER          ANALYST            106 19-APR-81
    3000              20

  EMPNO  ENAME          JOB               MGR HIREDATE
-----
    SAL      COMM      DEPTNO
-----
    103 TURNER          SALESMAN           105 08-SEP-81
    1500      0          30
    104 SMITHKO          CLERK              106 17-DEC-80
    800       0          20
    105 BLAKE            MANAGER            101 01-MAY-81
    2850      0          30

  EMPNO  ENAME          JOB               MGR HIREDATE
-----
    SAL      COMM      DEPTNO
-----
    106 JONES            MANAGER            101 02-APR-81
```



```

2975          20
107 ALLEN          SALESMAN          105 20-FEB-81
1600          300          30
108 WARDEN          SALESMAN          105 22-FEB-81
1250          500          30

EMPNO ENAME          JOB          MGR HIREDATE
-----
SAL          COMM          DEPTNO
-----
109 MARTINKING          SALESMAN          105 28-SEP-81
1250          1400          30
110 JAMES          CLERK          105 03-DEC-81
950          30
111 MILLERROBIN          CLERK          101 23-JAN-82
1300          10

EMPNO ENAME          JOB          MGR HIREDATE
-----
SAL          COMM          DEPTNO
-----
112 CLARKEN          MANAGER          101 09-JUN-81
2450          10
114 SARALA          CLERK          112 15-AUG-81
2100          10
115 RAMESH          ANALYST          112 10-OCT-81
2600          20

15 rows selected.

```

25. In WIPRO User issue a ROLLBACK and now check in SCOTT user.

SQL>rollback;

```

SQL> rollback;
Rollback complete.

```

26. ROLLBACK all the transactions in SCOTT and WIPRO Users.

SQL>rollback;

```

SQL> rollback;
Rollback complete.

```